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Few-Shot Sketch Recognition Framework

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Introduction

Few-shot sketch recognition trains systems to identify new object classes from only a few example sketches.

Sketches are abstract, sparse, and stylistically diverse, requiring specialized models to handle variability and limited training examples effectively.

Our High Level Approach

Our framework combines self-supervised learning with few-shot classification to overcome data scarcity in sketch recognition.

Stage 1: Self-Supervised Pre-training

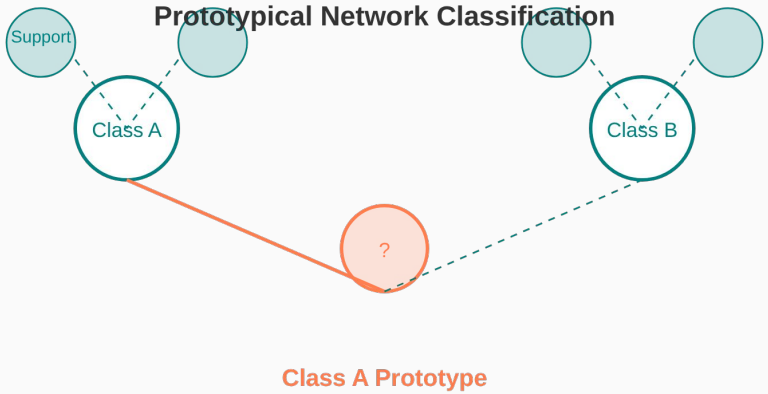
- Representation learning without labels
- Contrastive learning (SimCLR, MoCo)
- Non-contrastive learning (BYOL)
- Optimized embedding space

Stage 2: Few-Shot Classification

- Adapting pre-trained representations
- N-way K-shot learning
- Prototypical Networks
- Generalizing to novel classes

N-way K-shot Learning

N-way K-shot learning is a few-shot learning framework where N denotes the number of classes and K the number of labeled examples per class used for training or evaluation episodes.



Key Benefits:

- Leverages pre-trained representations
- Requires minimal fine-tuning

Sketch-Specific Data Augmentation

Effective data augmentation is crucial for self-supervised learning on sketches to generate diverse views while preserving semantic content.

Stroke Removal/Dropout

Randomly removes a percentage of strokes or segments to encourage learning from incomplete information.

Deformation/Jittering

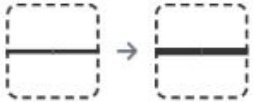
Applies slight non-rigid transformations to simulate variations in drawing style or perspective.

Affine Transformations

Rotation, scaling, translation, and shearing to introduce variability in orientation and size.

Color/Width Variation

Varying stroke width or inverting colors to create additional augmented views.



How to use of Dataset

Datasets

QuickDraw

- Over **50 million sketches**
- Across **345 Categories**
- Diverse collection of sketches

TU-Berlin

- **250 distinct** object classes
- **80 sketches** per class
- Fine-grained categories

Performance Metrics : Mean top-1 accuracy with 95% confidence intervals is reported over around 600 episodes to ensure statistical reliability and performance stability.

Uses for Pre-Training

Uses for Fine Tuning and testing

N-way
Novel classes

→

K-shot
Support examples

→

Query
Classification

Implementation Plan

1

Preprocessing and Augmentation

Convert sketches to grayscale bitmaps or vector strokes, resize to 224×224 pixels, and apply augmentations like scaling, rotation, elastic warping, stroke variation, erasing, and horizontal flipping to enhance model robustness.

2

Self-Supervised Pretraining

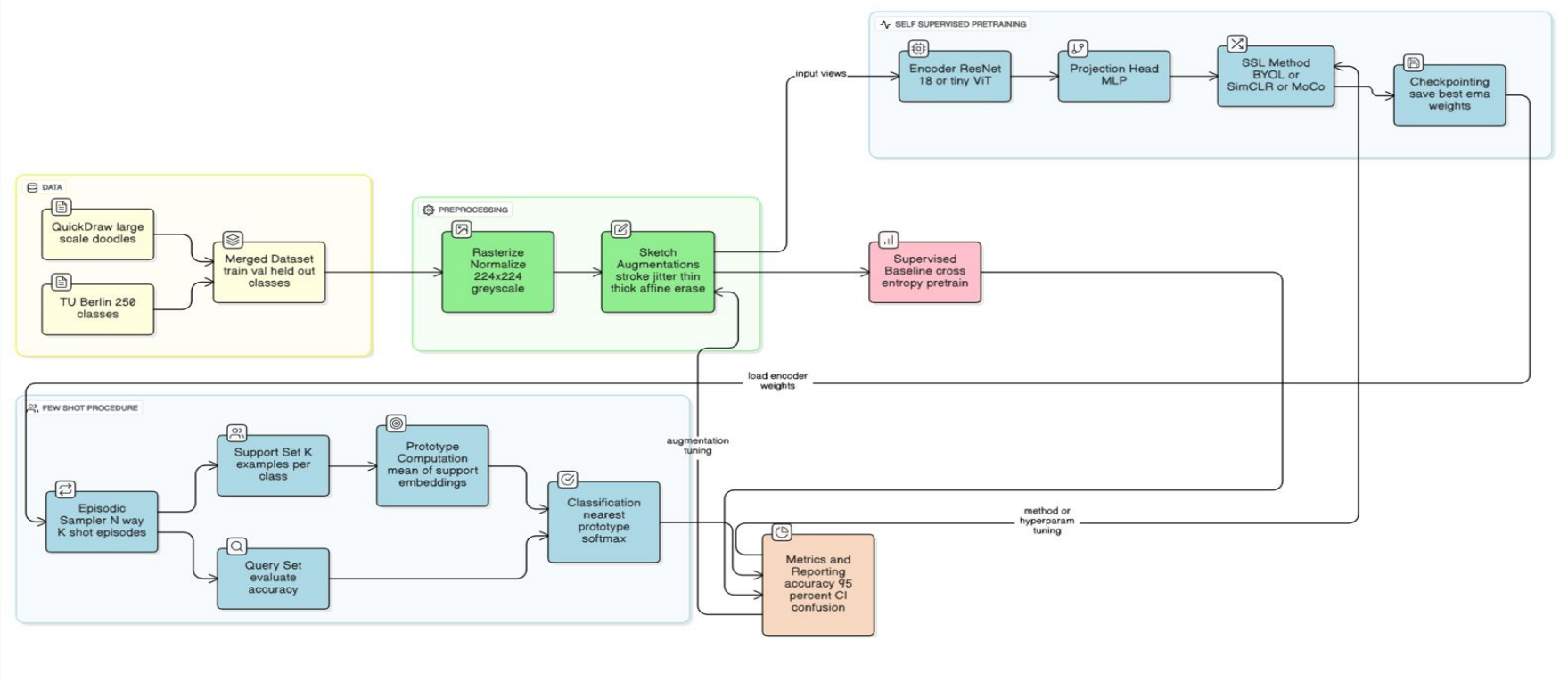
Use SSL methods such as BYOL or SimCLR with encoders like ResNet-18/34 or Sketch-a-Net, attach a projection head for 512-dimensional embeddings, and train on large sketch datasets to obtain pretrained models for downstream tasks.

3

Few-Shot Training and Evaluation

Implement linear-probe or episodic fine-tuning strategies for few-shot classification, evaluate on standardized tasks with multiple episodes, and compare against supervised and random baselines to assess generalization.

Flow Diagram



Project Timeline

Current | Future

Phase 2 (2 weeks)	Phase 3 (2 weeks)	Phase 4 (3 weeks)	Phase 5 (1 week)

Phase 1: Literature Review & Setup (Completed)

- Review of existing literature and project setup

Phase 2: Self-Supervised Model Implementation (2 weeks)

- Implementation of chosen self-supervised methods (e.g., SimCLR, BYOL) and pre-training

Phase 3: Few-Shot Framework Integration(2 weeks)

- Integration of pre-trained encoder with few-shot learning algorithms (e.g., Prototypical Networks)

Phase 4: Experiments & Analysis(3 weeks)

- Evaluation on TU-Berlin and QuickDraw datasets, comparison with baseline methods

Phase 5: Final Report & Presentation(1 week)

- Documentation of results, analysis, and presentation of the project outcomes

References

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